

# Foundation Models for Precision Classification in NSBI: toward a Higgs self-coupling ( $\kappa_\lambda$ ) measurement in $HH \rightarrow b\bar{b}\tau\tau$

*working note — draft for discussion*

19 June 2026

## Bottom line

For constraining  $\kappa_\lambda$ , the statistically-limited piece is *kinematic shape* discrimination of nearly-identical distributions. Both our own tests and the literature indicate this piece is **method-independent and statistics-limited**: a foundation model (FM) will *not* out-classify a good high-level-feature NSBI on the  $\kappa_\lambda$  shape (*though a systematics-robust FM can still help the  $\kappa_\lambda$  inference under profiling — see §6*). The genuine leverage of an FM is elsewhere — (i) **data efficiency** at low signal stats, (ii) **signal-vs-background acceptance/purity**, and (iii) **robustness to systematics**. The recommended path is a *factorized, calibration-first* strategy: keep the validated signal-extraction machinery and do NSBI on high-level kinematics now; pursue a jet-free / event-level FM on the acceptance axis as a staged, calibration-gated upgrade.

## 1 Physics goal and current status

We aim to constrain the Higgs trilinear self-coupling modifier  $\kappa_\lambda = \lambda_{HHH}/\lambda_{HHH}^{\text{SM}}$  through non-resonant di-Higgs production in the  $HH \rightarrow b\bar{b}\tau\tau$  final state, with the dominant sensitivity in the **resolved regime** ( $m_{HH} \in [250, 400]$  GeV).  $\kappa_\lambda$  enters as a shape distortion of  $m_{HH}$  (interference between the triangle,  $\propto \lambda_{HHH}$ , and box amplitudes in ggF).

A neutral-simulation-based-inference (NSBI) fit using **10 event-level features**,

$$m_{HH}, \cos\theta^*, p_T^{HH}, m_{bb}, \Delta R_{bb}, m_{\tau\tau}, \Delta\phi_{HH}, \Delta R_{\tau\tau}, p_T^{H_1}, p_T^{H_2},$$

already **tightens the  $\kappa_\lambda$  95% CI by roughly a factor of two** (with  $t\bar{t}$  background only, so far). The central *challenge* is statistical: very low signal yield, made worse once systematic uncertainties are profiled.

**Hypothesis under test:** a foundation-model + fine-tuning paradigm for the density-ratio (precision-classification) tasks inside the NSBI could improve the constraint — primarily via data efficiency and better object use.

## 2 The key decomposition: a $\kappa_\lambda$ NSBI is two ML jobs

The single most clarifying point is that the measurement contains two distinct learning problems, which an FM helps very differently:

The  $\kappa_\lambda$  *constraint* comes from job (B). Job (A) sets the effective signal purity and is where the analysis is acceptance- and systematics-limited. A flavor-pretrained FM is built for (A), not (B) — which explains the results below.

| Job                                    | Discriminates                                               | Information needed / best tool                                                                                                                  |
|----------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| (A) Signal vs. background              | $HH \rightarrow b\bar{b}\tau\tau$ vs. $t\bar{t}$ /QCD/fakes | <i>Object identity</i> : b-tagging, real-vs-fake $\tau$ , substructure — <b>flavor FM (Sophon-ak4) territory</b> .                              |
| (B) $\kappa_\lambda$ shape / inference | $\kappa_\lambda = 0$ vs. 5 vs. ... (morphing)               | <i>Event kinematics</i> ( $m_{HH}$ shape) — high-level features / NSBI; a “precision classification of nearly-identical distributions” problem. |

### 3 What we tested, and what we found

We adapted the `nsbi-lhc-toolkit` with a vendored Particle Transformer (ParT) backbone and **Sophon-ak4** (the AK4/resolved-jet ParT foundation model, pretrained for jet-flavor tagging on JetClass-II). A hierarchical model encodes each jet’s constituents and aggregates jets+objects with an event set-transformer. Three controls were compared on the  $\kappa_\lambda=0$  **vs.**  $\kappa_\lambda=5$  task (with  $\kappa_\lambda=5$  as the fixed reference/denominator): `scratch` (random-init ParT), `sophon_frozen`, `sophon_finetune`.

Table 1: \*  
Validation weighted-BCE (lower = better; random floor =  $\ln 2 = 0.693$ ).

| $N$ / point               | scratch       | sophon_finetune | sophon_frozen |
|---------------------------|---------------|-----------------|---------------|
| 100k (40 ep) <sup>†</sup> | 0.6378        | <b>0.6351</b>   | 0.6486        |
| 2,000                     | <b>0.6752</b> | 0.6775          | 0.6798        |
| 5,000                     | <b>0.6387</b> | 0.6457          | 0.6538        |

<sup>†</sup>earlier configuration (before early-stopping and the lowered fine-tune LR); single seed at low  $N$  (error bars pending).

#### Empirical finding

On the  $\kappa_\lambda$  *shape* task, pretraining does **not** help — the ordering is consistently **scratch**  $\leq$  **finetune**  $<$  **frozen**. Frozen flavor features are the worst; fine-tuning merely relearns toward scratch.

**Interpretation.** (i) *Pretext mismatch*: Sophon-ak4 is optimized for jet flavor, whose per-jet embeddings are lossy about the  $p_T$ /mass needed to reconstruct  $m_{HH}$  from cross-jet correlations. (ii) The task is precisely “precision classification of nearly-identical distributions,” which (next section) is statistics-limited and largely architecture-independent.

### 4 External evidence and the FM landscape

A 2025 review [1] organizes HEP foundation models into *supervised* (OmniLearn, Sophon, RS3L) and *self-supervised* (OmniJet- $\alpha$ , MPM) families. Points most relevant here:

- **Explicit vs. implicit priors are a wash — and statistics-limited** [2]. A direct comparison of L-GATr (Lorentz-equivariant) vs. OmniLearn (FM pretrained on 100M events) on *precision classification of similar classes* finds “comparable performance ... given the statistical precision of the datasets,” and that the gains from encoding physics are “method-independent.” This benchmark *is* our  $\kappa_\lambda$ -shape task, and predicts our null result. Nuances:

on the  $H1/ep$  task the FM *beat* the equivariant net, and for small signal injections the implicit (FM) approach was slightly better — **a mild lean toward FMs in the low-statistics corner.**

- **Taus are not a blocker.** OmniJet- $\alpha$  fine-tuned for hadronic- $\tau$  reconstruction improved momentum resolution by  $\sim 50\%$  vs. from-scratch training [3] — **so an earlier worry that OmniLearn-style FMs are unsuited to  $\tau$  final states was premature.**
- **Event-level FMs exist.** PECM [4] is a GNN over event objects (jets,  $e$ ,  $\mu$ ,  $\gamma$ , MET with four-momenta + type IDs) pretrained on 120M events; it reports  $>5$  percentage-point accuracy gains at  $10^3$ – $10^5$  events/class (our regime) and  $\sim 12\times$  faster fine-tuning, transferring across simulators. Closest match to our event-level inputs.
- **Self-supervision retains kinematics; RS3L targets systematics.** MPM [5] learns task-agnostic representations (less biased than supervised flavor labels). RS3L [6] pretrains on *re-simulation* augmentations (varied shower/hadronization/detector), yielding embeddings *robust to those systematics* — directly relevant to a systematics-limited measurement.

## 5 Two architectural strategies for the measurement

**Strategy A — jet-free, end-to-end.** Feed *all* particle-flow candidates of the event (no jet clustering) into a ParT/event-FM, add the 10 high-level features, and do NSBI for  $\kappa_\lambda$  in one model. Motivated by a recent *jet-free*  $HH \rightarrow 4b$  analysis [7]: ingesting all PF candidates ( $\sim 300$ /event) into a ParT gave  $>5\times$  search significance and an order-of-magnitude more retained signal at high background rejection. Caveats: it is *not* an FM (task-specific, trained on  $\sim 10^8$  events — “training statistics play a critical role”), it is  $HH \rightarrow 4b$  (a worse case for jet methods than  $b\bar{b}\tau\tau$ ), and it required a *new* “calibrate–validate–measure” scheme (calibrate on  $ZZ \rightarrow 4b$ , validate on  $ZH \rightarrow 4b$ ) because object-level scale factors do not apply to a jet-free discriminant.

**Strategy B — factorized.** Use the current CMS-style signal extractor for job (A), then NSBI on the high-level features (or an FM embedding) for job (B). The CMS  $HH \rightarrow b\bar{b}\tau\tau$  analysis [8] already uses a sophisticated DNN: a Lorentz Boost Network front-end, a  $\tau\tau$  neutrino-regression subnet, and a DenseNet classifier, with inputs that are full-system four-momenta *plus* ParticleNet/U-ParT (b-tag, c-vs-b, c-vs-l) and HHbTag scores — i.e. it already uses ParT as fixed tagger *scores* on reconstructed objects, with the entire validated systematics machinery built around those objects.

|                              | <b>A: jet-free FM (+10 feats) <math>\rightarrow</math> NSBI</b> | <b>B: CMS S/B <math>\rightarrow</math> NSBI on feats/FM</b>     |
|------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| Information ceiling          | Highest (soft radiation, no clustering/WP losses)               | High, capped by reconstruction                                  |
| Where it wins                | <b>Signal acceptance &amp; purity</b>                           | Clean $\kappa_\lambda$ -kinematics (the proven $\sim 2\times$ ) |
| $\kappa_\lambda$ -shape info | Same (kinematic)                                                | Same (kinematic)                                                |
| Calibration / systematics    | <b>Hard &amp; novel</b> (new scheme)                            | <b>Validated CMS machinery</b>                                  |
| NSBI tractability            | High-dim, harder closure/morphing                               | Low-dim, interpretable, easy closure                            |
| Data efficiency              | Needs huge stats unless pretrained                              | Already efficient                                               |
| Risk / time-to-result        | High / frontier                                                 | Low / near-term                                                 |

**Important caveat:** “Sophon-ak4 jet-free” **mixes scales** — Sophon-ak4 was pretrained on *single AK4 jets* ( $\sim 128$  constituents), so applying it to the *whole event* ( $\sim 300$  particles) is out-of-distribution. Jet-free-whole-event calls for an *event-scale* FM (PECM-style, or a from-scratch / SSL event-ParT), not Sophon-ak4 stretched beyond its domain.

## 6 Where the FM actually helps: separation vs. systematics robustness

A natural but *over-simplified* reading of the two-job decomposition is “the FM belongs in the  $s/b$  ratio, not the  $\kappa_\lambda$  ratio.” That holds for one benefit axis and fails for the other — and the one it fails for is precisely the systematics-limited regime that constrains this measurement. An NSBI  $\kappa_\lambda$  fit needs (at least) two learned ratios: a signal-vs-background ratio  $s(x)/b(x)$  and a  $\kappa_\lambda$ -parametrized signal-shape ratio  $r(x; \kappa_\lambda)$ . An FM is a *backbone for a ratio estimator*, so the question is always “which ratio, and helping *what*?”

- **Axis 1 — separation / central sensitivity.** How well a ratio discriminates (the kl0-vs-kl5 ceiling above). Here the 10 features are at ceiling and data-efficient, so an FM backbone does *not* improve the  $\kappa_\lambda$  ratio’s discrimination; its separation gain accrues only to  $s/b$  (acceptance, object ID).
- **Axis 2 — robustness + estimator stability under limited stats.** A property of the learned *representation*, not of its separation power. It can improve *either* ratio — including  $\kappa_\lambda$  — and it is the axis that matters when, as here, **the measurement is systematics- and statistics-limited.**

|                              | Axis 1: separation                   | Axis 2: robustness / stats stability                              |
|------------------------------|--------------------------------------|-------------------------------------------------------------------|
| $s/b$ ratio                  | FM helps (acceptance, object ID)     | FM helps (robust + data-efficient)                                |
| $\kappa_\lambda$ shape ratio | FM does <i>not</i> help (at ceiling) | FM <i>can</i> help (robust rep $\rightarrow$ smaller shape syst.) |

### How an FM helps the $\kappa_\lambda$ ratio on Axis 2:

1. *Robustness* (e.g. RS3L [6]): a representation invariant to shower/hadronization/detector variations yields a discriminant whose *shape* shifts less under nuisances  $\rightarrow$  **smaller profiled systematics  $\rightarrow$  tighter CI even at fixed separation.**
2. *Data-efficient systematic templates*: building systematic up/down variations needs ratios estimated on stats-starved varied MC; **a pretrained, data-efficient estimator gives less noisy templates  $\rightarrow$  better-behaved profiling.** (Largest where the estimator is **high-dimensional**; smaller for the already-efficient 10-feature ratio.)
3. (*Advanced*) *nuisance-parametrized NSBI*: learn  $r(x; \kappa_\lambda, \theta_{\text{syst}})$  directly; an FM backbone makes the higher-dimensional conditional learning more data-efficient.

### Catches (not free):

- To get the Axis-2 benefit on  $\kappa_\lambda$  you **build the ratio on the FM representation rather than the at-ceiling hand features** — trading interpretability / easy closure for robustness.
- You swap *well-calibrated object* systematics (JES,  $\tau$  energy scale) for the FM representation’s *own* systematics, which must then be established (the calibrate–validate–measure burden); robustness to known systematics can introduce new ones.
- Whether it is a *net* win is empirical — and it **will not appear in BCE/AUC**, which measure separation. We have so far measured only separation, which by construction cannot reveal a robustness gain.

### Corrected conclusion

The FM’s *separation* benefit accrues only to  $s/b$ ; its *robustness* / *data-efficiency* benefit can accrue to *either* ratio, including  $\kappa_\lambda$ . So an FM will not beat the high-level features on  $\kappa_\lambda$  *separation*, but **a systematics-robust / data-efficient FM (RS3L-style) can still help the  $\kappa_\lambda$  measurement by making the inference more robust under profiling when stats are limited** — the one place an FM earns its keep on the kinematic side.

**The diagnostic this implies.** The right test of the FM’s value here is *not* classifier loss but *robustness*: **propagate the dominant nuisances (JES,  $\tau$  energy scale, parton shower) through each control and measure** (i) how much the discriminant distribution shifts (nominal vs. varied) and (ii) **whether the profiled  $\kappa_\lambda$  CI shrinks**. A feature-based vs. FM-based comparison on *those* metrics — not BCE/AUC — is what would show an FM helping where the analysis is actually limited.

## 7 Synthesis and recommendation

### Key insights

1. The  $\kappa_\lambda$  constraint is **kinematic and statistics-limited**; explicit (equivariant) and implicit (FM) priors give comparable shape performance. Do not expect an FM to beat the high-level-feature NSBI on the  $\kappa_\lambda$  shape.
2. The FM’s real value is **data efficiency, S/B acceptance/purity, and systematics robustness** — not shape extraction.
3. For a systematics-limited result, **calibration of any learned discriminant is the dominant risk**; the jet-free upside can be eaten by a poorly understood systematic.
4. Use the right FM for the right job: Sophon-ak4 (jet-scale)  $\rightarrow$  per-jet S/B; event-scale FM  $\rightarrow$  jet-free acceptance; high-level features / NSBI  $\rightarrow \kappa_\lambda$ .

### Recommended staged plan.

- **Near-term (low risk):** Strategy B, sharpened. Do NSBI on the high-level features (optionally + the CMS DNN score) within the validated signal regions — the direct productization of the  $\sim 2\times$  result on real, calibrated data.
- **Test the right task:** a signal-vs- $t\bar{t}$  diagnostic (3 controls) — where per-jet Sophon *should* help, unlike the  $\kappa_\lambda$ -shape task. Plus a high-level kinematic ceiling baseline (BDT) to bound how much shape headroom exists.
- **Parallel R&D (high ceiling):** a jet-free / *event-level* FM for acceptance, introduced as a *calibratable* added discriminant (not a replacement), folded into the NSBI only once its calibration is validated.
- **Distinctive contribution:** RS3L-style re-simulation pretraining for a *systematics-robust* embedding — aimed squarely at the systematics bottleneck, and evaluated via the robustness diagnostic of §6 (nuisance shape variation + profiled CI), not BCE/AUC.

## 8 Open questions

- Do explicit and implicit priors learn the *same* physics, and does combining them (a Lorentz-equivariant *foundation* model) help in our regime?
- How large is the jet-free acceptance gain for  $b\bar{b}\tau\tau$  (cleaner than  $4b$ ), and can it be calibrated with  $Z \rightarrow \tau\tau/Z \rightarrow b\bar{b}$  control samples?
- Does any S/B or acceptance gain actually translate into a tighter  $\kappa_\lambda$  CI once systematics are profiled (the figure of merit is the constraint, not BCE/AUC)?

## References

- [1] *Foundation models for high-energy physics* (review), arXiv:2509.21434.
- [2] T. Wamorkar *et al.*, *Explicit or Implicit? Encoding Physics at the Precision Frontier*, arXiv:2603.08802.
- [3] *Reconstructing hadronically decaying tau leptons with a jet foundation model* (OmniJet- $\alpha$ ), arXiv:2503.19165, SciPost Phys. Core (2025).
- [4] *Pretrained Event Classification Model for High Energy Physics Analysis*, arXiv:2412.10665.
- [5] *Masked Particle Modeling on Sets: Towards Self-Supervised HEP Foundation Models*, arXiv:2401.13537.
- [6] *Re-Simulation-based Self-Supervised Learning for Pre-Training Foundation Models*, arXiv:2403.07066; Phys. Rev. D **111**, 032010 (2025).
- [7] *A calibratable, jet-free framework for  $HH \rightarrow 4b$  via all-particle multiclass classification*, arXiv:2508.15048.
- [8] CMS Collaboration, *Search for non-resonant Higgs boson pair production in the  $b\bar{b}\tau\tau$  final state with Run-3 data*, CMS AN-25-103 (draft, 2026).
- [9] O. Amram, K. Mikuni, B. Nachman *et al.*, *OmniLearn; OmniJet- $\alpha$* , arXiv:2403.05618.
- [10] *A Lorentz-Equivariant Transformer for All of the LHC* (L-GATr), arXiv:2411.00446; *Lorentz-Equivariant Geometric Algebra Transformers*, arXiv:2405.14806.